作业二

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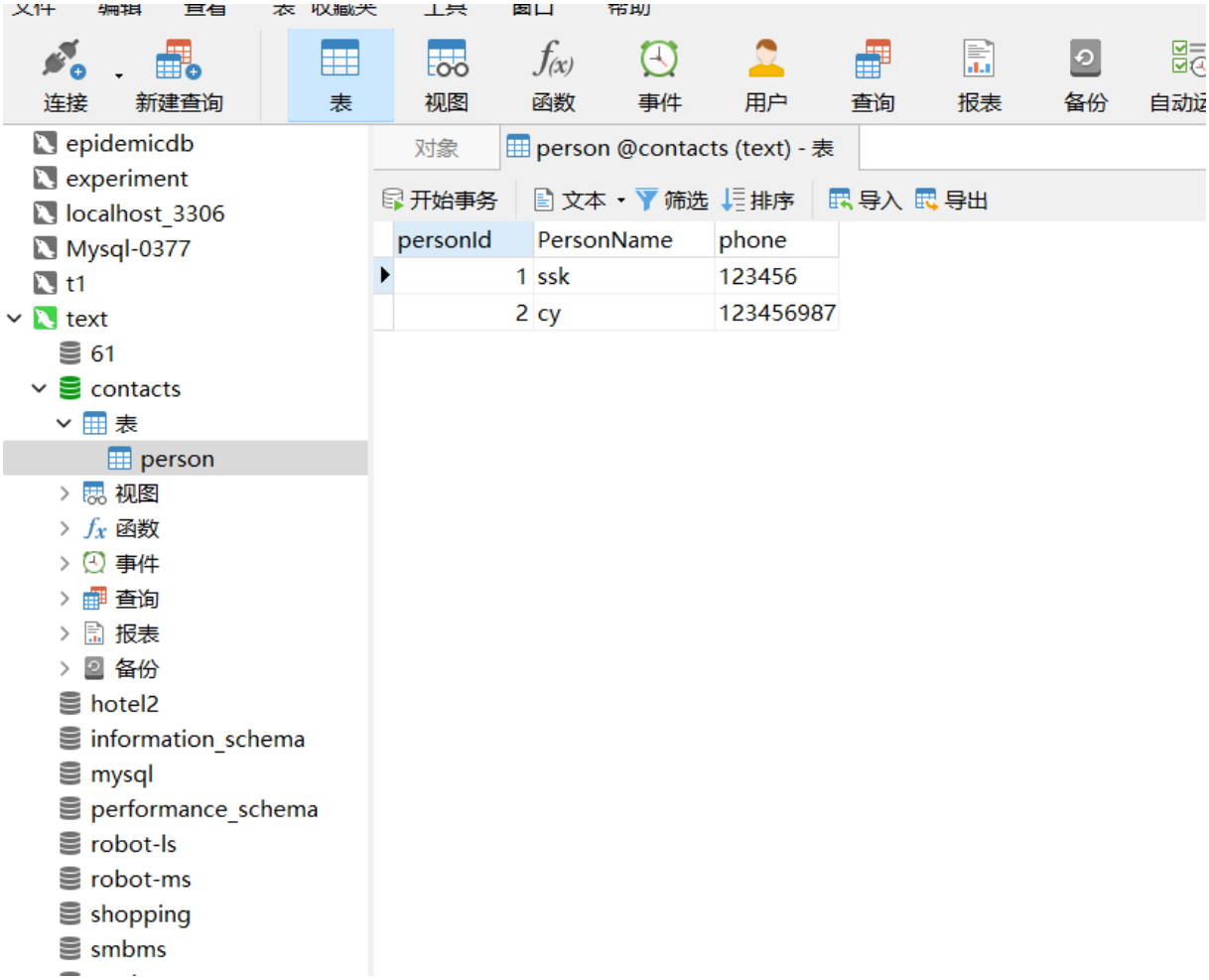
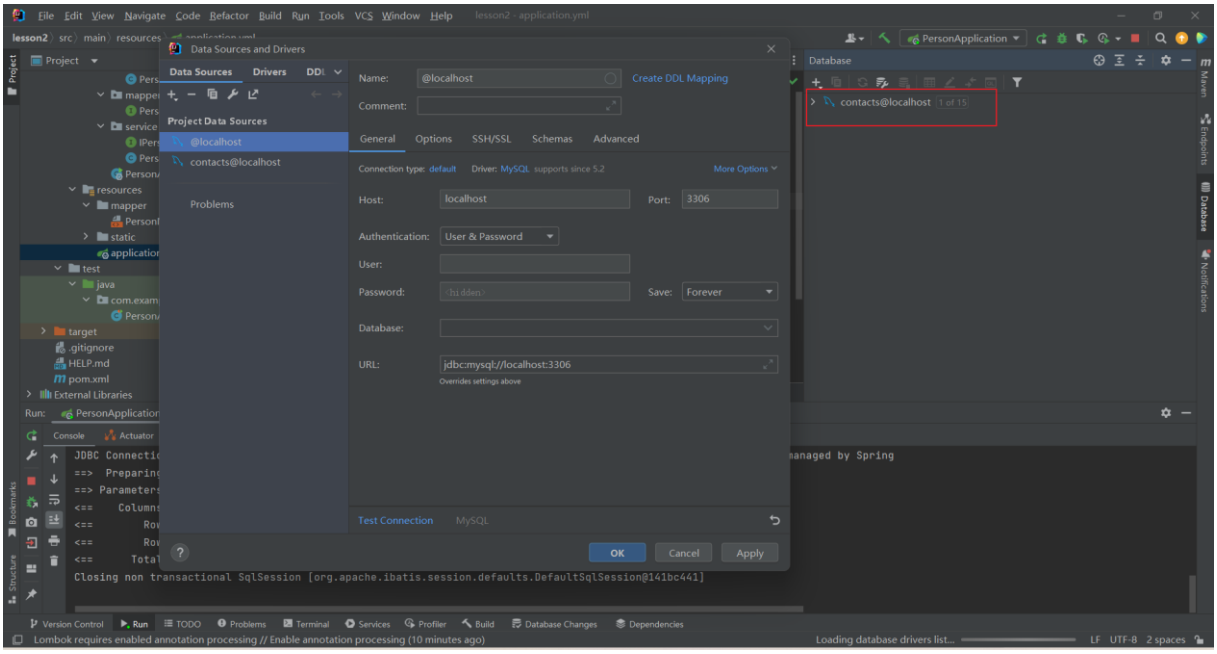
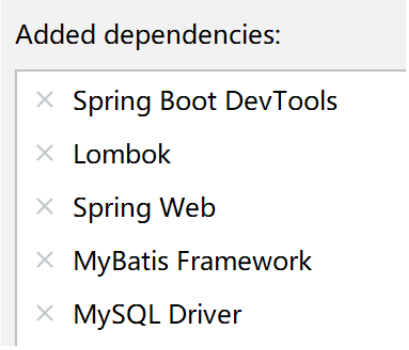
一、　mybatis 查询

1.1. 新建数据库，并创建表





1.2. 新建 Spring Boot 项目



3.pom.xml：添加依赖如下

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.2.2</version>
        <relativePath/> <!-- lookup parent from repository -->
    </parent>

    <groupId>com.example</groupId>
    <artifactId>springBootSelectDemo</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <name>springBootSelectDemo</name>
    <description>springBootSelectDemo</description>

    <properties>
        <java.version>17</java.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>

        <dependency>
            <groupId>org.mybatis.spring.boot</groupId>
            <artifactId>mybatis-spring-boot-starter</artifactId>
            <version>3.0.3</version>
        </dependency>

        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-devtools</artifactId>
            <scope>runtime</scope>
            <optional>true</optional>
        </dependency>
```

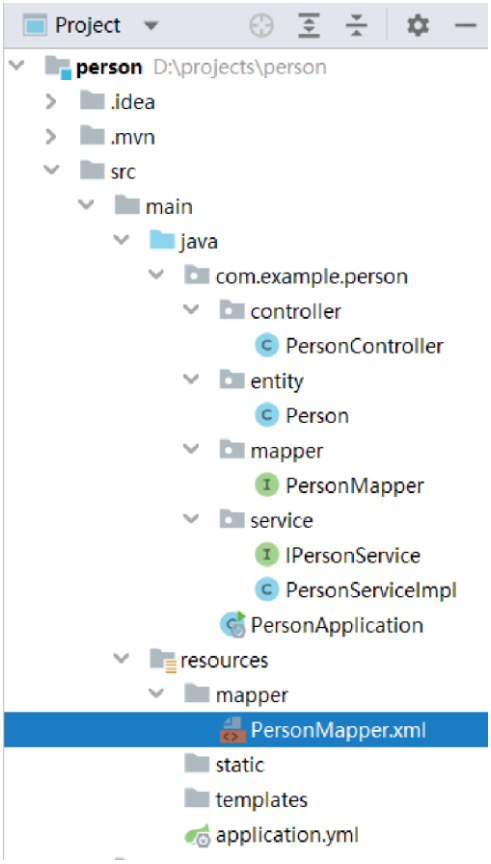
application.yml

```yaml
server:
  port: 80
spring:
  application:
    name: person
  datasource:
    driver-class-name: com.mysql.cj.jdbc.Driver
    url: jdbc:mysql://localhost:3306/contacts
    username: root
    password: root
mybatis:
  mapper-locations: classpath*:mapper/*Mapper.xml
  configuration:
    map-underscore-to-camel-case: true
    log-impl: org.apache.ibatis.logging.stdout.StdOutImpl
  type-aliases-package: com.example.person.entity
```

自动生成的 PersonApplication.java

```
package com.example.person;

import org.springframework.boot.SpringApplication;
import
org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class PersonApplication {

    public static void main(String[] args) {
        SpringApplication.run(PersonApplication.class, args);
    }

}
```

分别创建包 entity、mapper、service、controller 和文件



Person.java

```
package com.example.person.entity;

import lombok.Data;

@Data
public class Person {
    private Integer personId;
    private String personName;
    private String phone;
}
```

PersonMapper.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<!DOCTYPE mapper
        PUBLIC "-//mybatis.org//DTD Mapper 3.0//EN"
        "http://mybatis.org/dtd/mybatis-3-mapper.dtd">
<mapper namespace="com.example.person.mapper.PersonMapper">
    <select id="getPersonList" resultType="Person">
        SELECT * FROM person
    </select>
</mapper>
```

测试 PersonApplicationTests.java

```
package com.example.person;

import com.example.person.mapper.PersonMapper;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
class PersonApplicationTests {
    @Autowired
    private PersonMapper personMapper;

    @Test
    void testGetPersonList() {

personMapper.getPersonList().forEach(System.out::println);
    }

    @Test
    void contextLoads() {
    }

}
```

IPersonService.java

```
package com.example.person.service;

import com.example.person.entity.Person;

import java.util.List;

public interface IPersonService {
    List<Person> getPersonList();
}
```

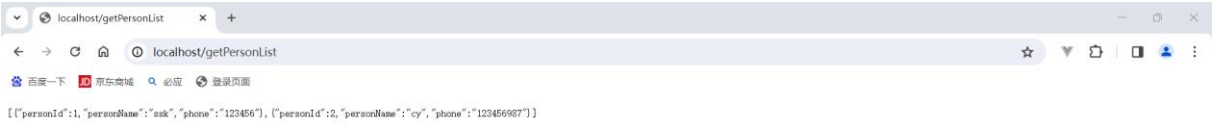
PersonServiceImpl.java

```
package com.example.person.service;

import com.example.person.entity.Person;
import com.example.person.mapper.PersonMapper;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import java.util.List;

@Service
public class PersonServiceImpl implements IPersonService{
    @Autowired
    private PersonMapper personMapper;

    @Override
    public List<Person> getPersonList() {
        return personMapper.getPersonList();
    }
}
```

PersonController.java

```
package com.example.person.controller;

import com.example.person.entity.Person;
import com.example.person.service.PersonServiceImpl;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
import org.springframework.web.bind.annotation.RestController;

import java.util.List;

@RestController
public class PersonController {
    @Autowired
    private PersonServiceImpl personService;

    @RequestMapping(value = "/getPersonList", method =
RequestMethod.GET)
    public List<Person> getPersonList() {
        return personService.getPersonList();
    }
}
```

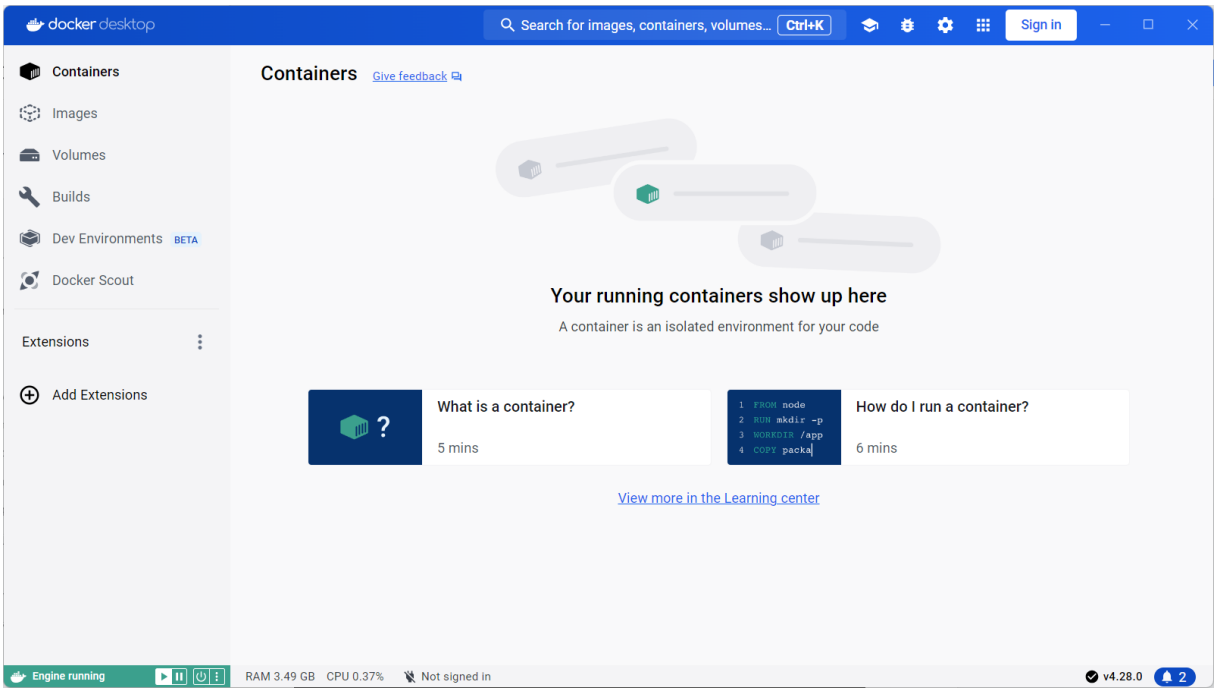
运行结果



代码分享：https://gitee.com/zhang-jike-kang/lesson2.git

二、 docker 安装 mysql

1. 下载 docker



2. 下载 wsl

```
sunshuaikang@DESKTOP-MF(

Windows PowerShell
版权所有 (C)  Microsoft Corporation。保留所有权利。

安装最新的 PowerShell，了解新功能和改进！ https://aka.ms/PSWindows

PS C:\Users\dell> wsl --install
Ubuntu 已安装。
正在启动 Ubuntu...
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: sunshuaikang
New password:
Retype new password:
passwd: password updated successfully
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.146.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage


This message is shown once a day. To disable it please create the
/home/sunshuaikang/.hushlogin file.
sunshuaikang@DESKTOP-MFGHONG:~$
```

3. 下载 jdk

```
sunshuaikang@DESKTOP-MF(

This message is shown once a day. To disable it please create the
/home/sunshuaikang/.hushlogin file.
sunshuaikang@DESKTOP-MFGHONG:~$ sudo apt-get update
[sudo] password for sunshuaikang:
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [1286 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [109 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [229 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [1591 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [267 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [851 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [162 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [16.8 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [37.1 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [7476 B]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [260 B]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:21 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [1502 kB]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [289 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [1619 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [271 kB]
Get:25 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1059 kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [240 kB]
```

```
sunshuaikang@DESKTOP-MFG
ol) in auto mode
Setting up humanity-icon-theme (0.6.16) ...
Setting up libgail18:amd64 (2.24.33-2ubuntu2) ...
Setting up libgtk2.0-bin (2.24.33-2ubuntu2) ...
Setting up openjdk-8-jdk:amd64 (8u402-ga-2ubuntu1~22.04) ...
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/appletviewer to provide /usr/bin/appletviewer (appletvi
ewer) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in aut
o mode
Setting up libgail-common:amd64 (2.24.33-2ubuntu2) ...
Setting up ubuntu-mono (20.10-0ubuntu2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ca-certificates (20230311ubuntu0.22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...

done.
done.
Processing triggers for libglib2.0-0:amd64 (2.72.4-0ubuntu2.2) ...
Processing triggers for libc-bin (2.35-0ubuntu3.4) ...
/sbin/ldconfig.real: /usr/lib/wsl/lib/libcuda.so.1 is not a symbolic link

Setting up at-spi2-core (2.44.0-3) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.8+dfsg-1ubuntu0.2) ...
sunshuaikang@DESKTOP-MFGHONG:~$ java -version
openjdk version "1.8.0_402"
OpenJDK Runtime Environment (build 1.8.0_402-8u402-ga-2ubuntu1~22.04-b06)
OpenJDK 64-Bit Server VM (build 25.402-b06, mixed mode)
sunshuaikang@DESKTOP-MFGHONG:~$
```

4. 安装 git

```
sunshuaikang@DESKTOP-MFGHONG:~$ sudo apt-get install git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.34.1-1ubuntu1.10).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 84 not upgraded.
sunshuaikang@DESKTOP-MFGHONG:~$ git version
git version 2.34.1
```

5. 下载 mysql

```
sunshuaikang@DESKTOP-MFG
Created symlink /etc/systemd/system/multi-user.target.wants/mysql.service → /lib/systemd/system/mysql.service.
Setting up libcgi-pm-perl (4.54-1) ...
Setting up libhtml-template-perl (2.97-1.1) ...
Setting up mysql-server (8.0.36-0ubuntu0.22.04.1) ...
Setting up libcgi-fast-perl (1:2.15-1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.4) ...
/sbin/ldconfig.real: /usr/lib/wsl/lib/libcuda.so.1 is not a symbolic link

sunshuaikang@DESKTOP-MFGHONG:~$ sudo service mysql start
sunshuaikang@DESKTOP-MFGHONG:~$ mysql -u root -p
Enter password:
ERROR 1698 (28000): Access denied for user 'root'@'localhost'
sunshuaikang@DESKTOP-MFGHONG:~$ mysql -u root -p
Enter password:
ERROR 1698 (28000): Access denied for user 'root'@'localhost'
sunshuaikang@DESKTOP-MFGHONG:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.36-0ubuntu0.22.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```